CFG for 'concat' function

entry:

%0 = icmp eq i32 %num_args, 3, !dbg !15
br i1 %0, label %assert_end, label %assert_fail, !dbg !15, !prof !16

T F

assert_end:

%.not = icmp eq ptr %args, null, !dbg !15
br i1 %.not, label %assert_fail1, label %assert_end2, !dbg !15, !prof !20

T F

assert_fail:

%1 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
. . . !17

tail call void %1(ptr nonnull @.str.1, ptr nonnull @.str), !dbg !15
br label %common.ret, !dbg !15

assert_fail1:

%2 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
. . . !17

tail call void %2(ptr nonnull @.str.1, ptr nonnull @.str.2), !dbg !15
br label %common.ret, !dbg !15

assert_end2:

%var_A.type_index = load i32, ptr %args, align 4, !dbg !15
%var_A.type_index.fr = freeze i32 %var_A.type_index, !dbg !15
%3 = icmp sgt i32 %var_A.type_index.fr, 63, !dbg !15
br i1 %3, label %assert_end4, label %switch.early.test, !dbg !15

T F

common.ret:

%common.ret.op = phi i32 [-1, %assert_fail], [-1, %assert_fail1], [
. . . -1, %assert_fail3], [-1, %assert_fail5], [-1, %assert_fail7], [-1,
. . . %assert_fail9], [-1, %assert_fail11], [-1, %assert_fail13], [-1,
. . . %assert_fail15], [-1, %assert_fail17], [-1, %assert_fail19], [-1,
. . . %assert_fail21], [-1, %assert_fail23], [-1, %assert_fail25], [-1,
. . . %assert_fail27], [-1, %assert_fail29], [-1, %assert_fail31], [-1,
. . . %assert_fail33], [-1, %assert_fail35], [-1, %assert_fail39], [-1,
. . . %assert_fail41], [-1, %assert_fail43], [-1, %assert_fail45], [-1,
. . . %assert_fail47], [-1, %assert_fail49], [-1, %assert_fail51], [-1,
. . . %assert_fail53], [-1, %assert_fail57], [-1, %assert_fail59], [-1,
. . . %assert_fail61], [-1, %assert_fail63], [-1, %assert_fail65], [0,
. . . %assert_end66]
ret i32 %common.ret.op, !dbg !15

assert_end4:

%5 = getelementptr inbounds i8, ptr %args, i64 16, !dbg !15
%var_B.type_index = load i32, ptr %5, align 4, !dbg !15
%var_B.type_index.fr = freeze i32 %var_B.type_index, !dbg !15
%6 = icmp sgt i32 %var_B.type_index.fr, 63, !dbg !15
br i1 %6, label %assert_end6, label %switch.early.test67, !dbg !15

T F

switch.early.test:

switch i32 %var_A.type_index.fr, label %assert_fail3 [
i32 7, label %assert_end4
i32 4, label %assert_end4
i32 0, label %assert_end4
], !dbg !15

def 7 4 0

assert_end6:

%8 = getelementptr inbounds i8, ptr %args, i64 32, !dbg !15
%var_T_concat.type_index = load i32, ptr %8, align 4, !dbg !15
%var_T_concat.type_index.fr = freeze i32 %var_T_concat.type_index, !dbg !15
%9 = icmp sgt i32 %var_T_concat.type_index.fr, 63, !dbg !15
br i1 %9, label %assert_end8, label %switch.early.test68, !dbg !15

T F

switch.early.test67:

switch i32 %var_B.type_index.fr, label %assert_fail5 [
i32 7, label %assert_end6
i32 4, label %assert_end6
i32 0, label %assert_end6
], !dbg !15

def 7 4 0

assert_fail3:

%4 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
. . . !17

tail call void %4(ptr nonnull @.str.1, ptr nonnull @.str.3), !dbg !15
br label %common.ret, !dbg !15

assert_end8:

%11 = getelementptr inbounds i8, ptr %args, i64 8, !dbg !15
%12 = load ptr, ptr %11, align 8, !dbg !15
%13 = icmp eq i32 %var_A.type_index.fr, 72, !dbg !15
%var_A.idx = select i1 %13, i64 16, i64 0, !dbg !15
%var_A = getelementptr inbounds i8, ptr %12, i64 %var_A.idx, !dbg !15
%14 = getelementptr inbounds i8, ptr %args, i64 24, !dbg !15
%15 = load ptr, ptr %14, align 8, !dbg !15
%16 = icmp eq i32 %var_B.type_index.fr, 72, !dbg !15
%var_B.idx = select i1 %16, i64 16, i64 0, !dbg !15
%var_B = getelementptr inbounds i8, ptr %15, i64 %var_B.idx, !dbg !15
%17 = getelementptr inbounds i8, ptr %args, i64 40, !dbg !15
%18 = load ptr, ptr %17, align 8, !dbg !15
%19 = icmp eq i32 %var_T_concat.type_index.fr, 72, !dbg !15
%var_T_concat.idx = select i1 %19, i64 16, i64 0, !dbg !15
%var_T_concat = getelementptr inbounds i8, ptr %18, i64 %var_T_concat.idx,
. . . !dbg !15
%.not69 = icmp eq ptr %12, null, !dbg !15
br i1 %.not69, label %assert_fail9, label %assert_end10, !dbg !15, !prof !20

T F

switch.early.test68:

switch i32 %var_T_concat.type_index.fr, label %assert_fail7 [
i32 7, label %assert_end8
i32 4, label %assert_end8
i32 0, label %assert_end8
], !dbg !15

def 7 4 0

assert_fail5:

%7 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15, !tbaa
. . . !17

tail call void %7(ptr nonnull @.str.1, ptr nonnull @.str.4), !dbg !15
br label %common.ret, !dbg !15

assert_fail9:

%20 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %20(ptr nonnull @.str.1, ptr nonnull @.str.6), !dbg !15
br label %common.ret, !dbg !15

assert_end10:

%21 = getelementptr inbounds i8, ptr %var_A, i64 16, !dbg !15
%22 = load i32, ptr %21, align 4, !dbg !15
%23 = icmp eq i32 %22, 2, !dbg !15
br i1 %23, label %assert_end12, label %assert_fail11, !dbg !15, !prof !16

T F

assert_fail7:

%10 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %10(ptr nonnull @.str.1, ptr nonnull @.str.5), !dbg !15
br label %common.ret, !dbg !15

assert_end12:

%25 = getelementptr inbounds i8, ptr %var_A, i64 24, !dbg !15
%concat.var_A.shape = load ptr, ptr %25, align 8, !dbg !15
%26 = load i64, ptr %concat.var_A.shape, align 8, !dbg !15, !tbaa !30
%M = trunc i64 %26 to i32, !dbg !15
%27 = getelementptr inbounds i8, ptr %concat.var_A.shape, i64 8, !dbg !15
%28 = load i64, ptr %27, align 8, !dbg !15, !tbaa !41
%N = trunc i64 %28 to i32, !dbg !15
%29 = getelementptr inbounds i8, ptr %var_A, i64 32, !dbg !15
%concat.var_A.strides = load ptr, ptr %29, align 8, !dbg !15
%30 = getelementptr inbounds i8, ptr %var_A, i64 12, !dbg !15
%dev_id = load i32, ptr %30, align 4, !dbg !15
%A = load ptr, ptr %var_A, align 8, !dbg !15
call void @llvm.assume(i1 true) ["align"(ptr %A, i64 64)], !dbg !15
%.not70 = icmp eq ptr %15, null, !dbg !15
br i1 %.not70, label %assert_fail13, label %assert_end14, !dbg !15, !prof !20

T F

assert_fail11:

%24 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %24(ptr nonnull @.str.1, ptr nonnull @.str.7), !dbg !15
br label %common.ret, !dbg !15

assert_fail13:

%31 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %31(ptr nonnull @.str.1, ptr nonnull @.str.8), !dbg !15
br label %common.ret, !dbg !15

assert_end14:

%32 = getelementptr inbounds i8, ptr %var_B, i64 16, !dbg !15
%33 = load i32, ptr %32, align 4, !dbg !15
%34 = icmp eq i32 %33, 2, !dbg !15
br i1 %34, label %assert_end16, label %assert_fail15, !dbg !15, !prof !16

T F

assert_end16:

%36 = getelementptr inbounds i8, ptr %var_B, i64 24, !dbg !15
%concat.var_B.shape = load ptr, ptr %36, align 8, !dbg !15
%37 = getelementptr inbounds i8, ptr %var_B, i64 32, !dbg !15
%concat.var_B.strides = load ptr, ptr %37, align 8, !dbg !15
%B = load ptr, ptr %var_B, align 8, !dbg !15
call void @llvm.assume(i1 true) ["align"(ptr %B, i64 64)], !dbg !15
%.not71 = icmp eq ptr %18, null, !dbg !15
br i1 %.not71, label %assert_fail17, label %assert_end18, !dbg !15, !prof !20

T F

assert_fail15:

%35 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %35(ptr nonnull @.str.1, ptr nonnull @.str.9), !dbg !15
br label %common.ret, !dbg !15

assert_fail17:

%38 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %38(ptr nonnull @.str.1, ptr nonnull @.str.10), !dbg !15
br label %common.ret, !dbg !15

assert_end18:

%39 = getelementptr inbounds i8, ptr %var_T_concat, i64 16, !dbg !15
%40 = load i32, ptr %39, align 4, !dbg !15
%41 = icmp eq i32 %40, 2, !dbg !15
br i1 %41, label %assert_end20, label %assert_fail19, !dbg !15, !prof !16

T F

assert_end20:

%43 = getelementptr inbounds i8, ptr %var_T_concat, i64 24, !dbg !15
%concat.var_T_concat.shape = load ptr, ptr %43, align 8, !dbg !15
%44 = getelementptr inbounds i8, ptr %var_T_concat, i64 32, !dbg !15
%concat.var_T_concat.strides = load ptr, ptr %44, align 8, !dbg !15
%T_concat = load ptr, ptr %var_T_concat, align 8, !dbg !15
call void @llvm.assume(i1 true) ["align"(ptr %T_concat, i64 64)], !dbg !15
%45 = getelementptr inbounds i8, ptr %var_A, i64 22, !dbg !15
%46 = load i16, ptr %45, align 2, !dbg !15
%47 = icmp eq i16 %46, 1, !dbg !15
%48 = getelementptr inbounds i8, ptr %var_A, i64 21, !dbg !15
%49 = load i8, ptr %48, align 1, !dbg !15
%50 = icmp eq i8 %49, 32, !dbg !15
%51 = getelementptr inbounds i8, ptr %var_A, i64 20, !dbg !15
%52 = load i8, ptr %51, align 1, !dbg !15
%53 = icmp eq i8 %52, 2, !dbg !15
%54 = and i1 %50, %53, !dbg !15
%55 = and i1 %47, %54, !dbg !15
br i1 %55, label %assert_end22, label %assert_fail21, !dbg !15, !prof !16

T F

assert_fail19:

%42 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %42(ptr nonnull @.str.1, ptr nonnull @.str.11), !dbg !15
br label %common.ret, !dbg !15

assert_end22:

%.not72 = icmp eq ptr %concat.var_A.strides, null, !dbg !15
br i1 %.not72, label %if_end, label %if_then, !dbg !15, !prof !20

T F

assert_fail21:

%56 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %56(ptr nonnull @.str.1, ptr nonnull @.str.12), !dbg !15
br label %common.ret, !dbg !15

if_end:

%69 = getelementptr inbounds i8, ptr %var_A, i64 40, !dbg !15
%70 = load i64, ptr %69, align 8, !dbg !15
%71 = icmp eq i64 %70, 0, !dbg !15
br i1 %71, label %assert_end26, label %assert_fail25, !dbg !15, !prof !16

T F

if_then:

%57 = load i64, ptr %concat.var_A.strides, align 8, !dbg !15, !tbaa !55
%58 = trunc i64 %57 to i32, !dbg !15
%59 = icmp eq i32 %N, %58, !dbg !15
%60 = icmp eq i32 %M, 1, !dbg !15
%61 = or i1 %60, %59, !dbg !15
%62 = getelementptr inbounds i8, ptr %concat.var_A.strides, i64 8, !dbg !15
%63 = load i64, ptr %62, align 8, !dbg !15, !tbaa !65
%64 = and i64 %63, 4294967295, !dbg !15
%65 = icmp eq i64 %64, 1, !dbg !15
%66 = icmp eq i32 %N, 1, !dbg !15
%67 = or i1 %66, %65, !dbg !15
%68 = and i1 %61, %67, !dbg !15
br i1 %68, label %if_end, label %assert_fail23, !dbg !15, !prof !16

T F

assert_end26:

%74 = getelementptr inbounds i8, ptr %var_A, i64 8, !dbg !15
%75 = load i32, ptr %74, align 4, !dbg !15
%76 = icmp eq i32 %75, 1, !dbg !15
br i1 %76, label %assert_end28, label %assert_fail27, !dbg !15, !prof !16

T F

assert_fail25:

%73 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %73(ptr nonnull @.str.1, ptr nonnull @.str.14), !dbg !15
br label %common.ret, !dbg !15

assert_fail23:

%72 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %72(ptr nonnull @.str.1, ptr nonnull @.str.13), !dbg !15
br label %common.ret, !dbg !15

assert_end28:

%78 = icmp ne ptr %A, null, !dbg !15
%79 = mul nsw i32 %N, %M, !dbg !15
%80 = icmp eq i32 %79, 0, !dbg !15
%81 = or i1 %80, %78, !dbg !15
br i1 %81, label %assert_end30, label %assert_fail29, !dbg !15, !prof !16

T F

assert_fail27:

%77 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %77(ptr nonnull @.str.1, ptr nonnull @.str.15), !dbg !15
br label %common.ret, !dbg !15

assert_end30:

%83 = getelementptr inbounds i8, ptr %var_B, i64 22, !dbg !15
%84 = load i16, ptr %83, align 2, !dbg !15
%85 = icmp eq i16 %84, 1, !dbg !15
%86 = getelementptr inbounds i8, ptr %var_B, i64 21, !dbg !15
%87 = load i8, ptr %86, align 1, !dbg !15
%88 = icmp eq i8 %87, 32, !dbg !15
%89 = getelementptr inbounds i8, ptr %var_B, i64 20, !dbg !15
%90 = load i8, ptr %89, align 1, !dbg !15
%91 = icmp eq i8 %90, 2, !dbg !15
%92 = and i1 %88, %91, !dbg !15
%93 = and i1 %85, %92, !dbg !15
br i1 %93, label %assert_end32, label %assert_fail31, !dbg !15, !prof !16

T F

assert_fail29:

%82 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %82(ptr nonnull @.str.1, ptr nonnull @.str.16), !dbg !15
br label %common.ret, !dbg !15

assert_end32:

%95 = load i64, ptr %concat.var_B.shape, align 8, !dbg !15, !tbaa !67
%96 = trunc i64 %95 to i32, !dbg !15
%97 = icmp eq i32 %M, %96, !dbg !15
br i1 %97, label %assert_end34, label %assert_fail33, !dbg !15, !prof !16

T F

assert_fail31:

%94 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %94(ptr nonnull @.str.1, ptr nonnull @.str.17), !dbg !15
br label %common.ret, !dbg !15

assert_end34:

%99 = getelementptr inbounds i8, ptr %concat.var_B.shape, i64 8, !dbg !15
%100 = load i64, ptr %99, align 8, !dbg !15, !tbaa !77
%101 = trunc i64 %100 to i32, !dbg !15
%102 = icmp eq i32 %N, %101, !dbg !15
br i1 %102, label %assert_end36, label %assert_fail35, !dbg !15, !prof !16

T F

assert_fail33:

%98 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %98(ptr nonnull @.str.1, ptr nonnull @.str.18), !dbg !15
br label %common.ret, !dbg !15

assert_end36:

%.not73 = icmp eq ptr %concat.var_B.strides, null, !dbg !15
br i1 %.not73, label %if_end38, label %if_then37, !dbg !15, !prof !20

T F

assert_fail35:

%103 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %103(ptr nonnull @.str.1, ptr nonnull @.str.19), !dbg !15
br label %common.ret, !dbg !15

if_end38:

%116 = getelementptr inbounds i8, ptr %var_B, i64 40, !dbg !15
%117 = load i64, ptr %116, align 8, !dbg !15
%118 = icmp eq i64 %117, 0, !dbg !15
br i1 %118, label %assert_end42, label %assert_fail41, !dbg !15, !prof !16

T F

if_then37:

%104 = load i64, ptr %concat.var_B.strides, align 8, !dbg !15, !tbaa !79
%105 = trunc i64 %104 to i32, !dbg !15
%106 = icmp eq i32 %N, %105, !dbg !15
%107 = icmp eq i32 %M, 1, !dbg !15
%108 = or i1 %107, %106, !dbg !15
%109 = getelementptr inbounds i8, ptr %concat.var_B.strides, i64 8, !dbg !15
%110 = load i64, ptr %109, align 8, !dbg !15, !tbaa !89
%111 = and i64 %110, 4294967295, !dbg !15
%112 = icmp eq i64 %111, 1, !dbg !15
%113 = icmp eq i32 %N, 1, !dbg !15
%114 = or i1 %113, %112, !dbg !15
%115 = and i1 %108, %114, !dbg !15
br i1 %115, label %if_end38, label %assert_fail39, !dbg !15, !prof !16

T F

assert_end42:

%121 = getelementptr inbounds i8, ptr %var_B, i64 8, !dbg !15
%122 = load i32, ptr %121, align 4, !dbg !15
%123 = icmp eq i32 %122, 1, !dbg !15
br i1 %123, label %assert_end44, label %assert_fail43, !dbg !15, !prof !16

T F

assert_fail41:

%120 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %120(ptr nonnull @.str.1, ptr nonnull @.str.21), !dbg !15
br label %common.ret, !dbg !15

assert_fail39:

%119 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %119(ptr nonnull @.str.1, ptr nonnull @.str.20), !dbg !15
br label %common.ret, !dbg !15

assert_end44:

%125 = getelementptr inbounds i8, ptr %var_B, i64 12, !dbg !15
%126 = load i32, ptr %125, align 4, !dbg !15
%127 = icmp eq i32 %dev_id, %126, !dbg !15
br i1 %127, label %assert_end46, label %assert_fail45, !dbg !15, !prof !16

T F

assert_fail43:

%124 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %124(ptr nonnull @.str.1, ptr nonnull @.str.22), !dbg !15
br label %common.ret, !dbg !15

assert_end46:

%129 = icmp ne ptr %B, null, !dbg !15
%130 = or i1 %80, %129, !dbg !15
br i1 %130, label %assert_end48, label %assert_fail47, !dbg !15, !prof !16

T F

assert_fail45:

%128 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %128(ptr nonnull @.str.1, ptr nonnull @.str.23), !dbg !15
br label %common.ret, !dbg !15

assert_end48:

%132 = getelementptr inbounds i8, ptr %var_T_concat, i64 22, !dbg !15
%133 = load i16, ptr %132, align 2, !dbg !15
%134 = icmp eq i16 %133, 1, !dbg !15
%135 = getelementptr inbounds i8, ptr %var_T_concat, i64 21, !dbg !15
%136 = load i8, ptr %135, align 1, !dbg !15
%137 = icmp eq i8 %136, 32, !dbg !15
%138 = getelementptr inbounds i8, ptr %var_T_concat, i64 20, !dbg !15
%139 = load i8, ptr %138, align 1, !dbg !15
%140 = icmp eq i8 %139, 2, !dbg !15
%141 = and i1 %137, %140, !dbg !15
%142 = and i1 %134, %141, !dbg !15
br i1 %142, label %assert_end50, label %assert_fail49, !dbg !15, !prof !16

T F

assert_fail47:

%131 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %131(ptr nonnull @.str.1, ptr nonnull @.str.24), !dbg !15
br label %common.ret, !dbg !15

assert_end50:

%144 = shl nsw i32 %M, 1, !dbg !15
%145 = load i64, ptr %concat.var_T_concat.shape, align 8, !dbg !15, !tbaa !91
%146 = trunc i64 %145 to i32, !dbg !15
%147 = icmp eq i32 %144, %146, !dbg !15
br i1 %147, label %assert_end52, label %assert_fail51, !dbg !15, !prof !16

T F

assert_fail49:

%143 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %143(ptr nonnull @.str.1, ptr nonnull @.str.25), !dbg !15
br label %common.ret, !dbg !15

assert_end52:

%149 = getelementptr inbounds i8, ptr %concat.var_T_concat.shape, i64 8,
. . . !dbg !15
%150 = load i64, ptr %149, align 8, !dbg !15, !tbaa !101
%151 = trunc i64 %150 to i32, !dbg !15
%152 = icmp eq i32 %N, %151, !dbg !15
br i1 %152, label %assert_end54, label %assert_fail53, !dbg !15, !prof !16

T F

assert_fail51:

%148 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %148(ptr nonnull @.str.1, ptr nonnull @.str.26), !dbg !15
br label %common.ret, !dbg !15

assert_end54:

%.not74 = icmp eq ptr %concat.var_T_concat.strides, null, !dbg !15
br i1 %.not74, label %if_end56, label %if_then55, !dbg !15, !prof !20

T F

assert_fail53:

%153 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %153(ptr nonnull @.str.1, ptr nonnull @.str.27), !dbg !15
br label %common.ret, !dbg !15

if_end56:

%164 = getelementptr inbounds i8, ptr %var_T_concat, i64 40, !dbg !15
%165 = load i64, ptr %164, align 8, !dbg !15
%166 = icmp eq i64 %165, 0, !dbg !15
br i1 %166, label %assert_end60, label %assert_fail59, !dbg !15, !prof !16

T F

if_then55:

%154 = load i64, ptr %concat.var_T_concat.strides, align 8, !dbg !15,
. . . !tbaa !103
%155 = trunc i64 %154 to i32, !dbg !15
%156 = icmp eq i32 %N, %155, !dbg !15
%157 = getelementptr inbounds i8, ptr %concat.var_T_concat.strides, i64 8,
. . . !dbg !15
%158 = load i64, ptr %157, align 8, !dbg !15, !tbaa !113
%159 = and i64 %158, 4294967295, !dbg !15
%160 = icmp eq i64 %159, 1, !dbg !15
%161 = icmp eq i32 %N, 1, !dbg !15
%162 = or i1 %161, %160, !dbg !15
%163 = and i1 %156, %162, !dbg !15
br i1 %163, label %if_end56, label %assert_fail57, !dbg !15, !prof !16

T F

assert_end60:

%169 = getelementptr inbounds i8, ptr %var_T_concat, i64 8, !dbg !15
%170 = load i32, ptr %169, align 4, !dbg !15
%171 = icmp eq i32 %170, 1, !dbg !15
br i1 %171, label %assert_end62, label %assert_fail61, !dbg !15, !prof !16

T F

assert_fail59:

%168 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %168(ptr nonnull @.str.1, ptr nonnull @.str.29), !dbg !15
br label %common.ret, !dbg !15

assert_fail57:

%167 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %167(ptr nonnull @.str.1, ptr nonnull @.str.28), !dbg !15
br label %common.ret, !dbg !15

assert_end62:

%173 = getelementptr inbounds i8, ptr %var_T_concat, i64 12, !dbg !15
%174 = load i32, ptr %173, align 4, !dbg !15
%175 = icmp eq i32 %dev_id, %174, !dbg !15
br i1 %175, label %assert_end64, label %assert_fail63, !dbg !15, !prof !16

T F

assert_fail61:

%172 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %172(ptr nonnull @.str.1, ptr nonnull @.str.30), !dbg !15
br label %common.ret, !dbg !15

assert_end64:

%177 = icmp ne ptr %T_concat, null, !dbg !15
%178 = mul nsw i32 %144, %N, !dbg !15
%179 = icmp eq i32 %178, 0, !dbg !15
%180 = or i1 %179, %177, !dbg !15
br i1 %180, label %assert_end66, label %assert_fail65, !dbg !15, !prof !16

T F

assert_fail63:

%176 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %176(ptr nonnull @.str.1, ptr nonnull @.str.31), !dbg !15
br label %common.ret, !dbg !15

assert_end66:

tail call fastcc void @concat_compute_(i32 %M, i32 %N, ptr %T_concat, ptr
. . . %B, ptr %A), !dbg !15
br label %common.ret, !dbg !15

assert_fail65:

%181 = load ptr, ptr @__TVMFFIErrorSetRaisedByCStr, align 8, !dbg !15,
. . . !tbaa !17
tail call void %181(ptr nonnull @.str.1, ptr nonnull @.str.32), !dbg !15
br label %common.ret, !dbg !15